

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 2

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Topic

Momentum Conservation and Medium Stress

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Momentum Conservation and Medium Stress. State all assumptions and conventions.

$$\partial_\mu (T^{\mu\nu} + S^{\mu\nu}) = 0$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$P_\Omega^i = \int_\Omega (T^{0i} + S^{0i}) d^3x$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$S^{\mu\nu} = \varepsilon_1 A^{\mu\nu} + \varepsilon_2 B^{\mu\nu}$$

Show enough intermediate work that another student can reproduce the calculation.